Lecture Homework Week 06 - Tuesday Template

Student Information

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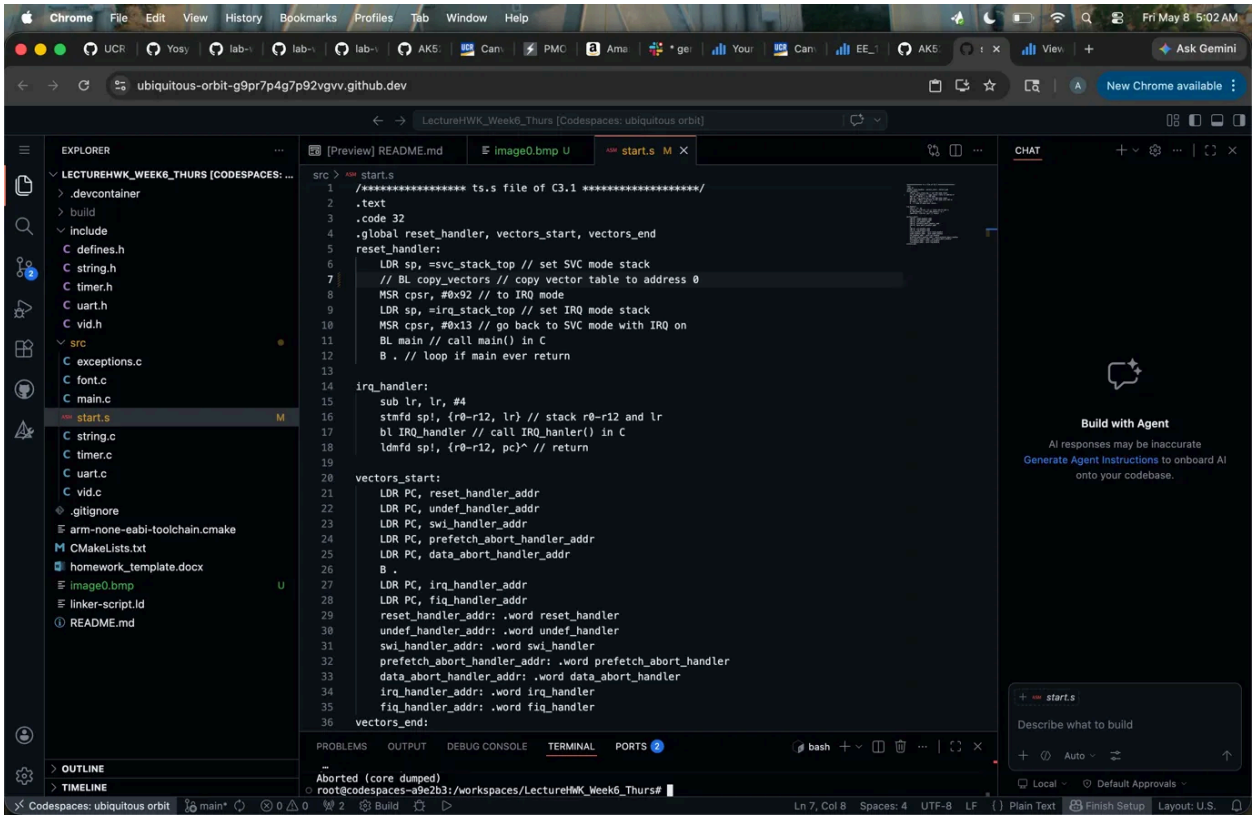
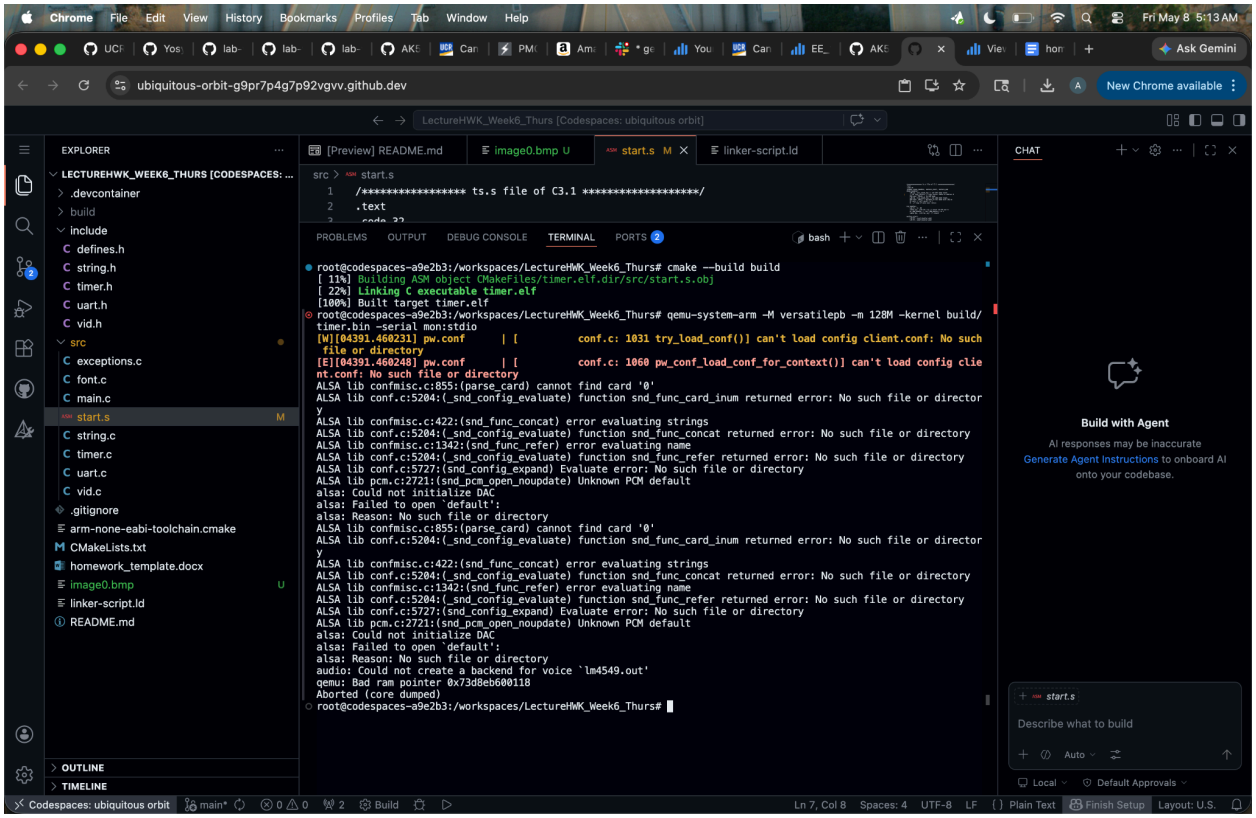
Student Responses

Answer the following questions, and be sure to highlight them when you turn them in on Gradescope

Problem 1

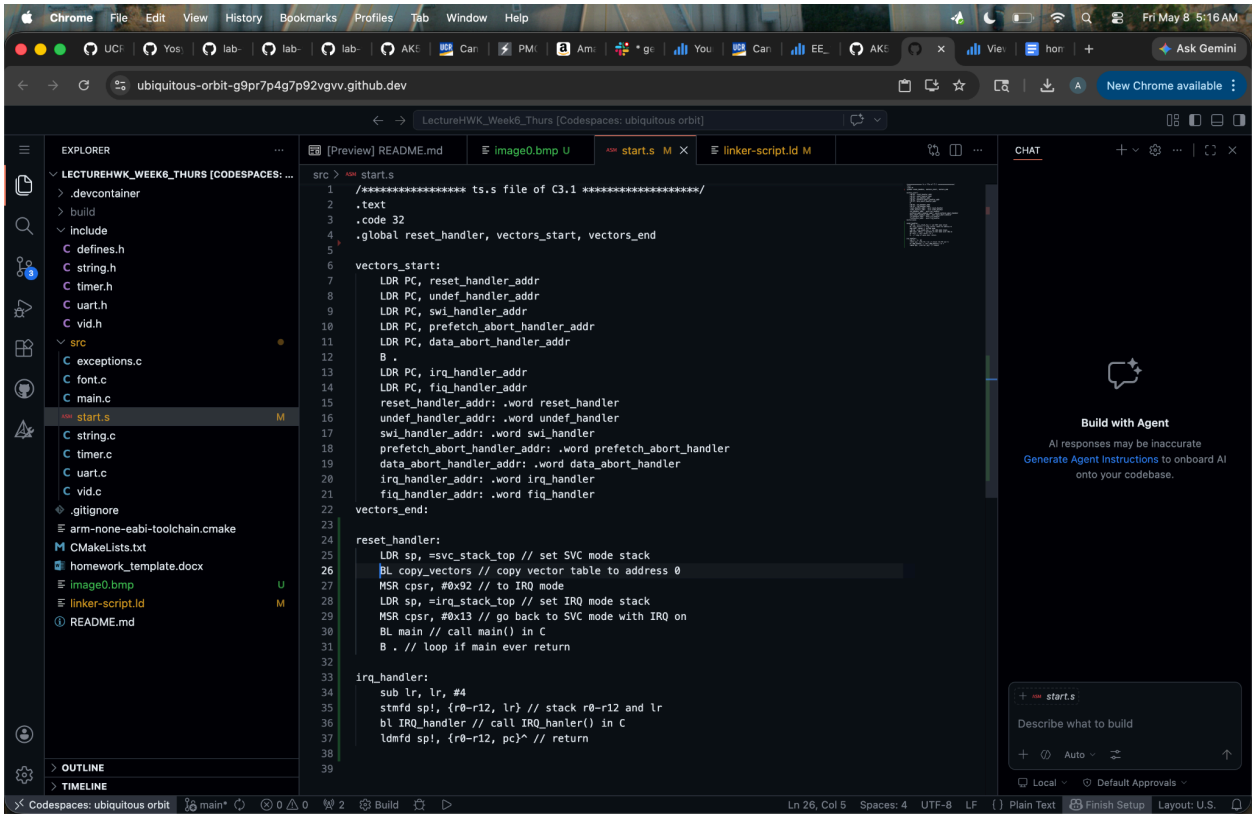
For part (1), below answer why the program does not work when you comment out the line `BL copy_vector:`

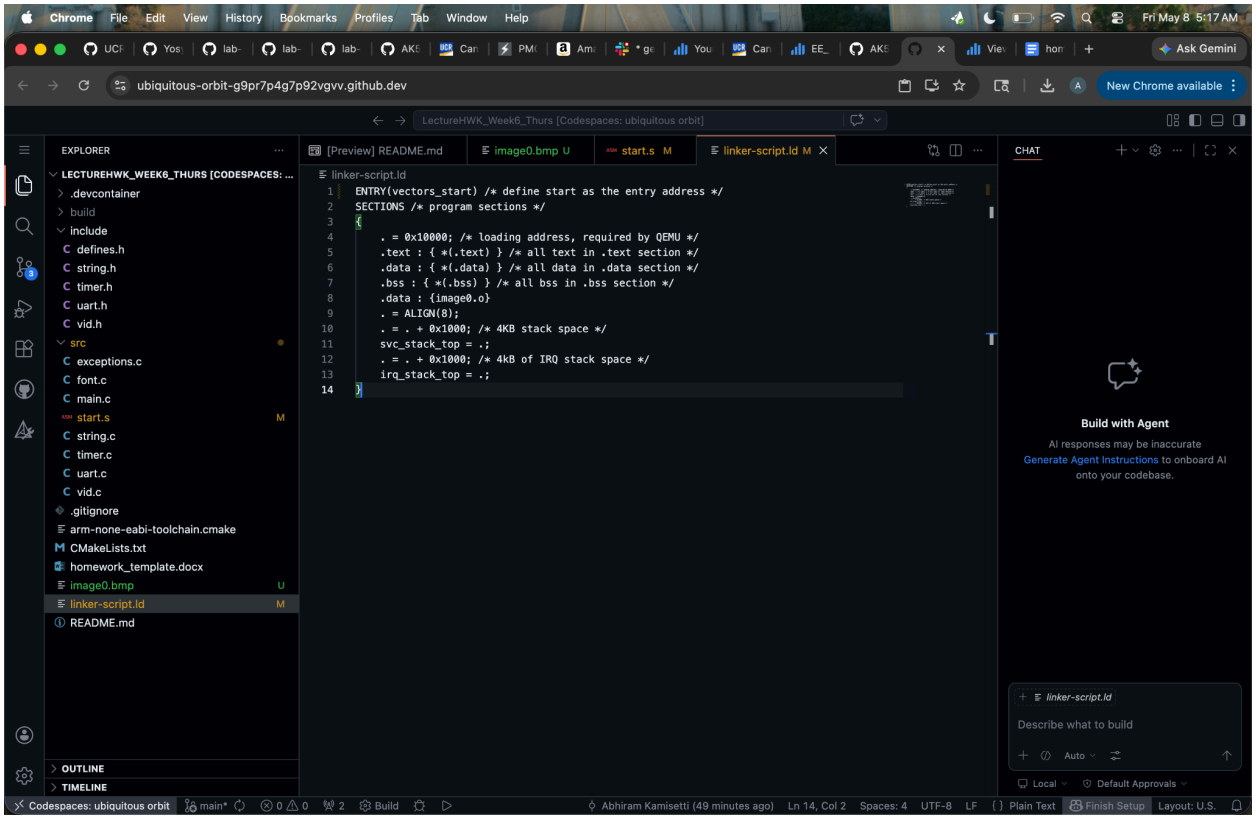




The ARM processor always checks 0x0 address when managing exceptions or interrupts. By commenting BL copy_vectors, the table is never copied from the 0x0 address so the processor discovers garbage values for data in the 0x0 address. This causes a bad memory address so it crashes

For part (2), below answer why moving the vector table to the beginning of the start.s file still works after recompiling:


ubiquitous-orbit-g9pr7p4g7p92vgvv.github.dev
EXPLORER
LECTUREHWK_WEEK6_THURS [CODESPACES: ...
[Preview] README.md
image0.bmp U
start.s M
linker-script.ld M
CHAT
/****************** ts.s file of C3.1 ******************/
.text
.code 32
.global reset_handler, vectors_start, vectors_end
vectors_start:
LDR PC, reset_handler_addr
LDR PC, undef_handler_addr
LDR PC, swi_handler_addr
LDR PC, prefetch_abort_handler_addr
LDR PC, data_abort_handler_addr
B .
LDR PC, irq_handler_addr
LDR PC, fiq_handler_addr
reset_handler_addr: .word reset_handler
undef_handler_addr: .word undef_handler
swi_handler_addr: .word swi_handler
prefetch_abort_handler_addr: .word prefetch_abort_handler
data_abort_handler_addr: .word data_abort_handler
irq_handler_addr: .word irq_handler
fiq_handler_addr: .word fiq_handler
vectors_end:
reset_handler:
LDR sp, =svc_stack_top // set SVC mode stack
BL copy_vectors // copy vector table to address 0
MSR cpsr, #0x92 // to IRQ mode
LDR sp, =irq_stack_top // set IRQ mode stack
MSR cpsr, #0x13 // go back to SVC mode with IRQ on
BL main // call main() in C
B . // loop if main ever return
irq_handler:
sub lr, lr, #4
stmfd sp!, {r0-r12, lr} // stack r0-r12 and lr
bl IRQ_handler // call IRQ_hanler() in C
ldmfd sp!, {r0-r12, pc}^ // return
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Codespaces: ubiquitous orbit
Ln 26, Col 5
Spaces: 4
UTF-8
LF
Plain Text


ubiquitous-orbit-g9pr7p4g7p92vgvv.github.dev
EXPLORER
LECTUREHWK_WEEK6_THURS [CODESPACES: ...
[Preview] README.md
image0.bmp U
start.s M
linker-script.ld M
CHAT
ENTRY(vectors_start) /* define start as the entry address */
SECTIONS /* program sections */
{
. = 0x10000; /* loading address, required by QEMU */
.text : { *(.text) } /* all text in .text section */
.data : { *(.data) } /* all data in .data section */
.bss : { *(.bss) } /* all bss in .bss section */
.data : {image0.o}
. = ALIGN(8);
. = . + 0x1000; /* 4KB stack space */
svc_stack_top = .;
. = . + 0x1000; /* 4kB of IRQ stack space */
irq_stack_top = .;
}
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Codespaces: ubiquitous orbit
Ln 14, Col 2
Spaces: 4
UTF-8
LF
Plain Text